

Swaroop Vaze

Computer

Engineering

Company: PhonePe

Job Profile: SDE

Offer Type: Internship + Placement

Internship Stipend: 80k

CTC: 33.5L (19.25L base + 5L joining bonus + 8.5L stocks)

Location: Bangalore

Process Date: - PPT + OA : 7th July, 2025

- Interview : 8th July, 2025

Placement Process:

1. [Online Assessment] [219 candidates appeared]

The assessment was conducted around 1hr after the ppt presentation (~4.30pm - 5pm). It was conducted on the DoSelect platform. The test contained 4 questions to be solved in 1hr 30 mins.

The webcam was on for the entire duration. There were also constraints on usb devices, so external keyboard and mouse may malfunction. Do ensure a proper laptop with a webcam, keyboard and trackpad. The college wifi was not working properly so some of us had to use our own mobile hotspot to give the test, so ensure you have adequate internet connection.

I personally faced lag during the contest where the text input was delayed 3-5s, I was writing entire 2-3 lines and the UI would update it seconds later. Also there was no VIM or EMACS mode for editing, so keep that in mind.

The running test suite and custom tests feature in this platform was abysmal.

- A. You are given 'n' people, labeled from 1 to n. There are 'm' existing friendships represented by 2 arrays 'A' and 'B', each of size 'm'. The i-th friendship is between person A[i] and person B[i].

Two people belong to the same friend group if they are direct friends or have some indirect friendship.

Find the minimum number of friendships to be added such that every person belongs to the same friend group.

Input:

The first line of each test case contains two integers n and m ($1 \leq n, m < 2 \cdot 10^5$)

The second line of each test case contains m integers, $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$)

The second line of each test case contains m integers, $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq n$)

Output:

A single integer - the minimum number of friendships to be added.

Example

input

4 2

1 3

2 4

Output

1

This can easily be solved by counting the number of connected components and returning the **number of connected components - 1**. I used DSU here and passed all test cases.

- B. You have ' n ' employees and have been given an array ' A ' of size ' n ', where $A[i]$ indicates the number of ladoos the i -th employee receives. If $A[i] = 0$, it means the ladoos are not yet assigned. You are also given 2 integers ' m ' and ' k '. Where ' m ' is the maximum number of ladoos a single employee can get. And ' k ' is the maximum allowed difference between adjacent employees.

Find the number of valid ways to assign ladoos to the unassigned positions such that. Each unassigned position gets a laddoo between 1 to m and the absolute difference between adjacent employees is at most k .

Input:

The first line of each test case contains two integers n, m and k ($1 \leq n \leq 10^4$)

($1 \leq m \leq 100$) ($1 \leq k \leq 30$)

The second line of each test case contains m integers, $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq m$)

Output:

A single integer - the total number of valid ways to assign the ladoos mod $1e9 + 7$

Example:

Input:

5 5 2
0 3 0 0 4

This was a simple 2D DP problem on counting, i just kept an array 'dp' of length 'm + 1', where dp[i] represents the number of ways to i ladoos currently. Calculated the dp by iterating over the n elements in array 'a'.

I passed most of the test cases except 2, where I didn't account for the first element being 0.

- C. You have been given an array 'a' of length 'n' and an integer 'k'. For each element the value of a[i] is defined as the sum of (bit * position) where bit is the bit at the position starting from 1 from LSB in the binary representation of a[i].

E.g. for a[i] = 13;

Binary representation of 13 = 1101 1 1 0 1

Position 4 3 2 1

Value = 4*1+3*1+2*0+1*1;

Find number of pairs(i ,j) such that value(a[i]) + value(a[j]) <= k

Input:

The first line of each test case contains two integers n and k ($1 \leq n \leq 10^5$)($1 \leq k \leq 10^5$)

The second line of each test case contains m integers, a1, a2, ... , an ($0 \leq a_i \leq 10^5$)

Output:

A single integer - the total number pairs(i ,j) such that value(a[i]) + value(a[j]) <= k

Input:

4 20

5 6 7 8

Output:

1

The expected solution for this problem was to make the value array first, and then using sorting + 2 pointers to find the number of pairs in ($O(n \cdot \log n)$). But I was running low on time so I wrote a brute force solution ($O(n^2)$) in 5 minutes and it passed all test cases.

- D. I don't remember this question, it was related to harry potter or something. The input format for this question was also not clear, so instead of wasting my time on it, I checked how the output was. We were supposed to print possible or impossible.

I just wrote `cout << "possible" << endl;` and it passed 6/8 test cases 🤖

In total I attempted 2 Complete + 2 partial questions, and I advanced to the next round. 30 students were shortlisted. The list of the students eligible for next rounds was released around 11 pm. And the next rounds were on the right next day.

2. **[Technical Round 1]** [30 candidates were shortlisted]

We were told to arrive at college at 8am, as the order of panels was not confirmed. But the interviews started at around 9am.

Problem 1: given an array of n integers return the maximum subarray sum of a subarray where the first element and last element is the same. Can be done by simple map in linear time.

<https://www.geeksforgeeks.org/dsa/maximum-sum-subarray-such-that-start-and-end-values-are-same/>

Problem 2: given a tree with all unique elements and queries of the form [currnode, value], you are supposed to output the maximum of (value xor node) over all nodes from root to currnode. This was solvable by a binary trie. Where you have to perform dfs and dynamically add and delete the visited nodes from the trie. And answer queries for currnode.

<https://codeforces.com/blog/entry/66434>

I did the first problem entirely and the second one with some hints. From every slot of panels 3 students were asked to give HM round first. From the 10 people in my panel, 3 students including me were asked to give HM round first. I wrote code for both the problems. And was expected to know the implementation of trie.

As time was running out, I did not ask any questions to the interviewer, but you should prepare some questions to ask them.

3. **[Hiring Manager Round]** [9(from directly tech round 1) + 5(after both tech rounds)]

This was a resume round, typical HR questions were asked. Conflicts with coworkers? How did you resolve it? Your weaknesses? How would you learn fintech?.

I was lucky enough to get asked about my 6 month internship and projects I did in it. My normal projects section only had a single project (NOT RECOMMENDED) about employee management systems, I was not asked a question regarding this. Do not depend on luck, have both good internships and projects.

He then asked me to give me my favourite feature any app has made in recent years. I told him about the apple natural handwriting to math feature and explained its design. This is very

common system design question and it is generally a good idea to be prepared in advance for this.

4. [Technical Round 2] [11 were shortlisted for the final round]

Some people who completed the HM round first were also rejected, so do not panic if you were not in the 9 people who were chosen to give HM round first.

The format of this round was the same as the Technical Round 1. But the difficulty of the questions was high.

Problem 1: given an array of strings return true if concatenation of any two strings makes a palindrome. This can be solved using trie. And I was again said to write the whole implementation of the trie.

<https://leetcode.com/problems/palindrome-pairs/description/>

Problem 2: this problem was 2700 on codeforces, in my entire codeforces run i have solved at max a 2500 question. I would not even explain this just go to the link below. This was just to test how the student thinks and I don't think they expected any of us to solve this question.

<https://codeforces.com/contest/1270/problem/G>

I solved the first problem without any hint. But I could not solve the second problem with any hints. My interview was the longest by far. Around 1hr 45 mins.

Out of 11 students 7 were selected, I was one of them.

Useful Tips:

1. DSA is required 100%. I did not touch DSA in the last 2 months before the interview because I was busy with my internship. But the intuition from 2 years of codeforces and leetcode came in handy. Non beginner concepts like binary lifting were also asked, so be prepared.
2. CP is not necessary it just helps, even 2 experts were disqualified in the HM round so It is not true that if you have CP skills only you will pass, having a good resume is must.
3. Do not put fake information in your resume - you will be asked about it in detail so you should know the ins and outs of it. Which will only be true if you actually put what you did and not some random github clone. Do not try to fill the resume with things you have half knowledge about.
4. Have good communication skills, most of the time the interview just wants to converse with you. There is no need to have top notch english, it is enough if you can convey your words in simple sentences.
5. Core subjects were not asked to me, but I advise to have good knowledge of those.
6. System design basics are required.